

Notes: Antràs, Garicano & Rossi-Hansberg (QJE, 2006)
Offshoring in a Knowledge Economy

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1 Big picture

- **Question.** How does the possibility of forming cross-country production teams change work organization, worker-manager matching, wage inequality, firm size, and trade?
- **Core mechanism.** Low-skill agents specialize in production; high-skill agents solve problems for many workers. Offshoring lets managers in one country leverage knowledge over workers in another country.
- **Main result.** Globalization improves matches for all southern workers, but only for the best northern workers. Southern worker inequality rises; northern worker inequality can rise or fall depending on communication cost h and skill overlap α .
- **Trade result.** The South exports physical goods; the North exports knowledge services. If knowledge services are not recorded, trade statistics can show a southern surplus and a northern deficit.

2 Incremental contribution of the paper

- It develops a hierarchical one-to-many matching model rather than one-to-one assignment.
- It endogenizes who becomes a worker and who becomes a manager.
- It derives skill complementarity from problem solving and knowledge specialization rather than imposing it directly.
- It makes communication technology a primitive that governs managerial span of control and therefore the organization of offshoring.
- It connects globalization to wage inequality, firm-size distributions, and trade in knowledge services in one model.

3 Model primitives and equations

Agents have skill z , which is the fraction of production problems they can solve. A manager with skill z_m supervises workers with skill z_p . Communication cost is h .

$$h(1 - z_p)n = 1, \quad n(z_p) = \frac{1}{h(1 - z_p)}.$$

Managers choose worker skill to maximize rents:

$$R(z_m) = \max_{z_p} \frac{z_m - w(z_p)}{h(1 - z_p)}, \quad w'(z_p) = \frac{z_m - w(z_p)}{1 - z_p}.$$

With assignment function $m(z)$, labor-market clearing implies

$$m'(z) = h(1 - z) \frac{g(z)}{g(m(z))}.$$

Interpretation. Positive sorting makes m increasing: better workers match with better managers. Since managers lead many workers, matching is many-to-one.

4 Closed and integrated equilibria

In the South, skills are uniform on $[0, \alpha]$; in the North, skills are uniform on $[0, 1]$. For the closed South,

$$m_S(z) = z_S^* + hz(1 - z/2), \quad z_S^* = \frac{1 + h - \sqrt{1 + h^2 + 2h(1 - \alpha)}}{h}.$$

Globalization allows cross-country teams. Depending on h and α , the integrated economy has a low-quality or high-quality offshoring equilibrium.

5 Identification and theoretical design

The argument is theoretical. Its credibility comes from showing that a unique and efficient closed-economy equilibrium exists, deriving the global matching functions, and comparing the integrated economy to autarky. The key assumptions are positive sorting, costly communication, exogenous skill distributions, and two-layer teams.

6 Figures and extracted theorem/proposition blocks

6.1 Theorem 1: closed economy equilibrium

Read: Theorem 1 establishes existence, uniqueness, positive sorting, convex earnings, and efficiency for sufficiently low h .

Economic message. This gives the paper a clean autarkic benchmark.

in z .

THEOREM 1. There exists a threshold $h^* > 0$ such that if $h \in [0, h^*]$ there exists a unique competitive equilibrium of this economy. In equilibrium the set of managers and the set of workers are connected, the equilibrium exhibits positive sorting, and the earnings function is strictly convex. Furthermore, the equilibrium allocation is efficient.

II.A. Equilibrium in the Closed Economy

Consider a world formed by two independent economies

Theorem 1: closed economy equilibrium properties

6.2 Figure I: Types of Equilibria

Read: The figure divides the (h, α) space into low-quality and high-quality offshoring equilibria.

Economic message. The qualitative effect of globalization depends jointly on communication technology and skill overlap.

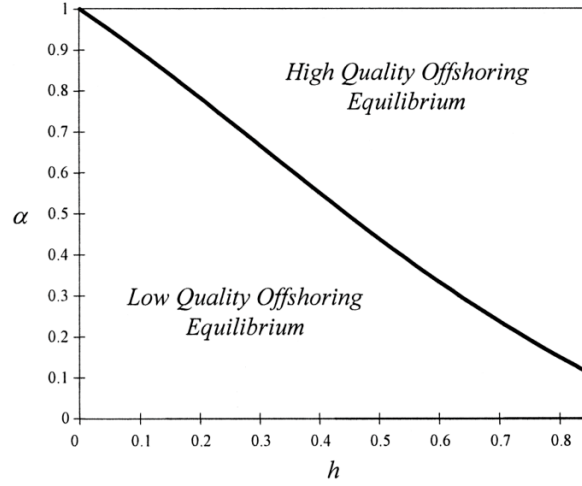


FIGURE I
Types of Equilibria

Figure I: Types of Equilibria

6.3 Figure II and Figure III: Low-Quality and High-Quality Offshoring Equilibria

Read: The figures show wages, rents, densities, and skill partitions in the two global equilibria.

Economic message. Offshoring is a reallocation of worker and manager skill intervals, not a simple relocation of one task.

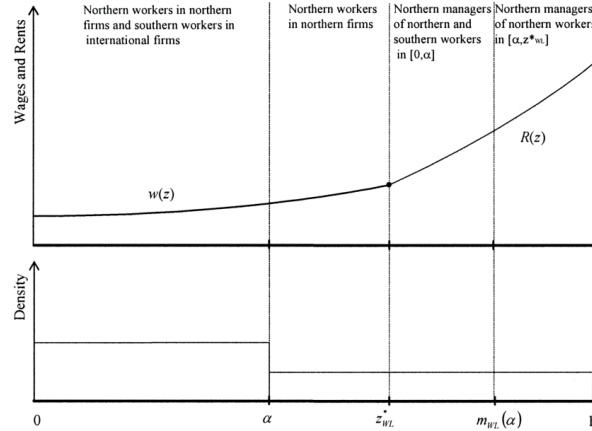


FIGURE II
Low Quality Offshoring Equilibrium

(a) Figure II: Low Quality Offshoring Equilibrium

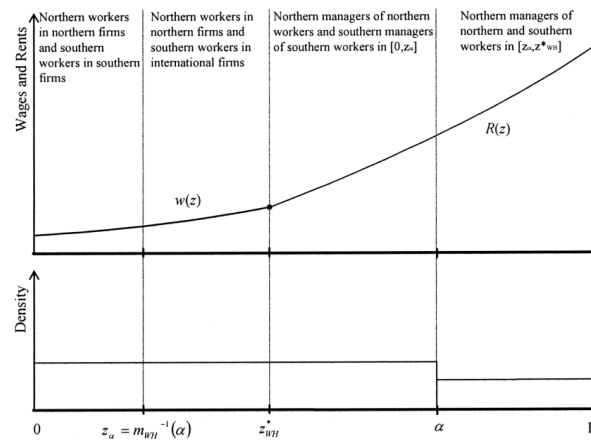


FIGURE III

(b) Figure III: High Quality Offshoring Equilibrium

6.4 Figure IVa and Figure IVb: Matching before and after globalization

Read: These figures compare autarkic matching with global matching in LQE and HQE.

Economic message. They show the central mechanism: southern workers improve their matches, while northern workers split into winners and losers.

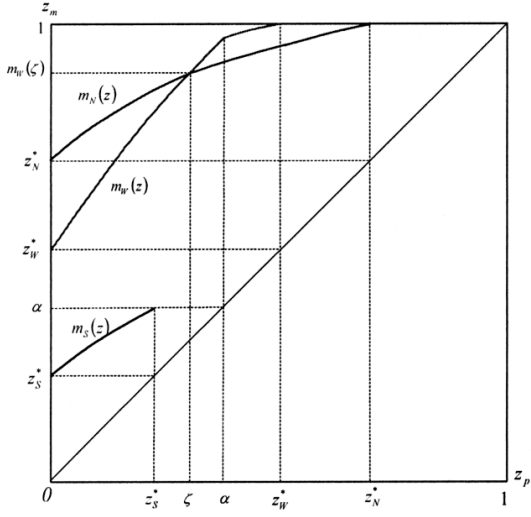


FIGURE IVa

(a) Figure IVa: Matching before and after globalization in LQE

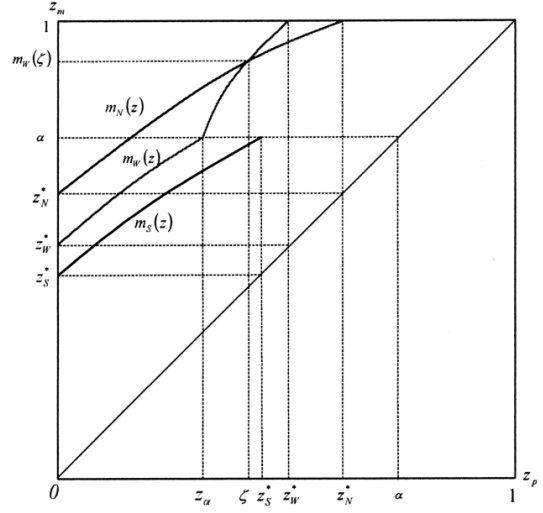


FIGURE IVb

atching before ($m_N(z)$, $m_W(z)$) and after ($m_N(z)$) globalization in a HQE

(b) Figure IVb: Matching before and after globalization in HQE

6.5 Proposition 1: Globalization and team formation

Read: Globalization raises southern worker mass and northern manager mass; all southern workers improve matches; northern workers below a threshold get worse matches and those above get better matches.

Economic message. The proposition explains the within-country distributional effects of offshoring.

- (i) The mass of southern workers and the mass of northern managers both increase, i.e., $z_S^* < z_W^* < z_N^*$;
- (ii) (a) Southern workers who were already workers are matched with a better manager;
- (b) Southern managers who remain managers are matched with worse workers;
- (c) Southern managers who become workers are matched with a northern manager;
- (iii) (a) There exists a unique threshold ζ such that all northern workers who remain workers with $z < \zeta$ are matched with a worse manager, while those with $z > \zeta$ are matched with a better manager;
- (b) All northern managers who were already managers with $z < m_W(\zeta)$ are matched with a better worker, while those with $z > m_W(\zeta)$ are matched with a worse worker.

Proposition 1: Globalization and team formation

6.6 Proposition 2, Summary 1, and Summary 2: Wage inequality and wage levels

Read: Southern worker inequality increases. Northern worker inequality increases when h and α are low, but decreases when they are high. Some low-skill agents lose.

Economic message. The model predicts asymmetric inequality effects across North and South.

SUMMARY 1. Globalization increases within-worker wage inequality in the North if h and α are sufficiently low, but it decreases it if h and α are sufficiently high. Furthermore, globalization decreases the marginal return to skill of all northern workers with knowledge z below a threshold, but increases the marginal return to skill of all northern workers above this threshold.

The analysis of the impact of globalization on inequality follows quite directly from the changes in matching. First, inequality within southern workers unambiguously increases. The marginal value of workers' skill is driven by the skill of the manager with whom they are matched, which increases for all southern workers. Thus, the sum of the complementarity and the

We are also interested in studying the effect of globalization on the level of wages, which is the result of two effects: on one hand, southern workers face more competition from low skill northern workers; on the other hand, globalization improves their match. In contrast, in the North, workers also face more competition, but they do not always improve their match (as described in Proposition 1). If we focus attention on the effect of globaliza-

(a) Proposition 2

(b) Summary 1–2

6.7 Summary 3: Production, consumption, and trade balance

Read: Globalization raises production in the South, lowers production in the North, raises consumption in both, and creates manufactures exports from South and knowledge-service exports from North.

Economic message. Unrecorded knowledge transfers can make trade statistics misleading.

added, the “winner” of globalization is the South. The reason why the North produces less after globalization is that physical production does not take into account that managers' rents have to be repatriated. Managers consume in their own country, and they receive—as compensation for the time spent helping and communicating with workers abroad—part of the production of these workers. These rents can be substantial and in fact imply that consumption in both countries increases and, since utility is linear, so does welfare.

This difference in consumption and production outcomes is reflected in the trade balance of these countries. In particular, the South features net exports of physical goods, while the North features net exports of knowledge services.²⁷ Furthermore, if

Summary 3: Production, consumption, and trade balance

6.8 Propositions 3–5: Comparative statics

Read: Proposition 3 studies the skill of the world's most-skilled worker; Proposition 4 studies the quantity and quality of offshoring; Proposition 5 studies communication costs, matching, and firm size.

Economic message. Communication improvements and southern skill upgrading affect both the quantity and the quality margins of offshoring.

Occupational Choice. As h decreases, managers have access to a communication technology that allows them to deal with larger teams. As a result, the number of workers in the South weakly decreases with h , and the number of workers in the North decreases with h . The number of managers in the South in turn weakly increases with h , and the number of managers in the North increases with h . As α increases, agents in the South

(a) Proposition 3

the quantity and quality of offshoring. To measure the quantity of offshoring, we need a measure that captures the extent to which firms in the economy are formed by national versus international teams. Therefore, we define the *quantity* of offshoring as the proportion of southern workers that work for international teams. In a LQE, all agents in the South are workers in international teams, so it follows that our measure of offshoring equals one. In contrast, in a HQE, the quantity of offshoring is given by

$$O_W \equiv (z_{WH}^* - z_\alpha^*)/z_{WH}^*.$$

(b) Proposition 4

ers, and will thus work for more productive firms. Workers with high skill levels will be matched with worse managers and so the productivity of the firms for which they work will decrease. We formalize these conclusions in the next proposition.

PROPOSITION 5. A decrease in communication costs (h) has the following effects on matching and firm size:

- (i) It improves the match for workers below a threshold skill level φ , while it worsens the match for workers (who were already workers) above φ ;
- (ii) It increases the size of all firms.

An increase in the skill overlap (α) has the following effects on matching and firm size:

Proposition 5

7 Short summary

- Offshoring is modeled as cross-country hierarchical team formation.
- Workers produce; managers solve problems; communication costs determine managerial span of control.
- Globalization improves all southern worker matches but creates winners and losers among northern workers.
- Southern worker inequality rises; northern worker inequality depends on h and α .
- The model explains firm creation/destruction, firm-size changes, and knowledge-service trade.

8 Conclusion

8.1 Core conclusion

Globalization changes the organization of production by rematching workers and managers across countries. It raises welfare in both countries but redistributes match quality and wage gains across skill groups.

8.2 Incremental contribution

The paper's incremental contribution is the integration of offshoring with hierarchical one-to-many matching and endogenous occupational choice. It explains wage inequality, firm size, and trade flows through one organizational mechanism.

8.3 Economic intuition

Managerial knowledge is scalable. Better communication lets high-skill northern managers combine their knowledge with southern routine labor. Southern workers gain better matches; some northern workers face worse matches; high-skill northern agents can leverage knowledge more widely.

8.4 Limitations and extensions

The paper abstracts from contracting frictions, firm boundaries, endogenous skill acquisition, self-employment, and detailed task heterogeneity. These are natural extensions for understanding multinational boundaries and human-capital responses.

8.5 Takeaway

Offshoring is an organizational shock, not merely a trade shock. It reshapes hierarchies, occupations, wage inequality, firm size, and measured trade.